

# Recurring decimals → fractions — past-paper practice

Four real Edexcel IGCSE Mathematics A questions (spec 4MA1), taken straight from the papers. Show full algebraic working — the “show that” questions need the algebra, not just the answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Time: \_\_\_\_\_

**Q1** 2019 Paper 1H, Q15

2 marks

Show that  $0.25\dot{4} = 14/55$

**15** Use algebra to show that the recurring decimal  $0.2\dot{5}\dot{4} = \frac{14}{55}$

(Total for Question 15 is 2 marks)

Show that  $0.28\dot{1}\dot{3} = \frac{557}{1980}$

16 Use algebra to show that the recurring decimal  $0.28\dot{1}\dot{3} = \frac{557}{1980}$

(Total for Question 16 is 2 marks)

**$0.4\dot{x}$  as a fraction, in terms of  $x$**

**18**  $0.4\dot{x}$  is a recurring decimal.

$x$  is a whole number such that  $1 \leq x \leq 9$

Find, in terms of  $x$ , the recurring decimal  $0.4\dot{x}$  as a fraction.

Give your fraction in its simplest form.

Show clear algebraic working.

.....  
(Total for Question 18 is 3 marks)

Show that  $0.\dot{6}\dot{1}\dot{2} = 101/165$

16 Use algebra to show that the recurring decimal  $0.\dot{6}\dot{1}\dot{2} = \frac{101}{165}$

(Total for Question 16 is 2 marks)

**Reminder:** 9s = one per digit in the repeating block · 0s = one per digit after the point before the block · top = digits up to the end of the first block – digits in front · then simplify. Answers are on the revision page under "Past-paper practice".